

Complex analysis

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Complex Numbers

complex numbers can be put into two parts

$$\operatorname{Re} z = x, \operatorname{Im} z = y$$

and let's say $z_i = (x_i, y_i)$, then

$$(i) \ z_1 = z_2 \iff x_1 = x_2, y_1 = y_2$$

$$(ii) \ z_1 + z_2 = (x_1 + x_2, y_1 + y_2)$$

$$(iii) \ z_1 z_2 = (x_1 x_2 - y_1 y_2, y_1 x_2 + x_1 y_2)$$

we know that z can be expressed in terms of $z = r(\cos \theta + i \sin \theta)$

and since $r = \sqrt{x^2 + y^2}$, we also know that $r = |z|$, showing $\tan \theta = y/x$.

the angle θ therefore is also called $\arg z$, where $\arg z = \operatorname{Arg} z + 2n\pi$

this also shows that $\arg z \equiv \operatorname{Arg} z \pmod{2\pi}$

now it's easy to prove de Moivre's formula

$$(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta$$

because

$$z = re^{i\theta}, z^n = r^n e^{in\theta}$$

and the exponent being multiplied by n means it comes into coefficient in front of θ inside $\sin$ or $\cos$.

Definition: it is called neighborhood if $D(z_0, \delta)$ is defined this way

$$D(z_0, \delta) = \{z \in C : |z - z_0| < \delta\}$$

open set is defined where

$$\forall z \in S, \exists \delta > 0 \text{ such that } D(z, \delta) \subset S$$

bounded means

$$\forall z, |z_n| \leq M$$

where converging means

$$\forall n \geq N \rightarrow |z_n - z| < \epsilon$$

the fair point is that there could be some functions that bounds but doesn't converge, such as this: $z_n = i^n$.

this is because if we set this as

$$\forall n > N \rightarrow |z_n - z| < 1/2$$

then $4N + 1 > N$ and $4N + 3 > N$ is valid therefore

$$2 = |i^{-4N+1} - z - (i^{-4N+3} - z)| \leq |z_{4N+1} - z| + |z_{4N+3} - z| < 1/2 + 1/2 = 1$$

so it doesn't prove.

definition it is said to be Cauchy sequence if

$$\forall m, n \geq N \rightarrow |z_m - z_n| < \epsilon$$

And here comes significant theorem,

Theorem Bolzano-Weierstrass

if S is bounded, infinite set, then S owns a limit. especially, if $\{z_n\}$ is unique complex sequence, bounded, meaning $\exists M > 0$,

$$|z_n| \leq M, \forall n = 1, 2, 3, \dots$$

then $\{z_n\}$ owns a converging partial sequence.

Analytic Function

limit is defined as

$$\forall 0 < |z - z_0| < \delta, \exists |f(z) - w_0| < \epsilon$$

Theorem Suppose

$$f(z) = u(x, y) + iv(x, y)$$

is differentiable at $z_0 = x_0 + iy_0$. Then u and v have first-order partial derivatives at (x_0, y_0) and satisfy the Cauchy-Riemann equations

$$u_x = v_y, \quad u_y = -v_x \quad (2.45)$$

Moreover, $f'(z_0)$ can be written as

$$f'(z_0) = u_x(x_0, y_0) + iv_x(x_0, y_0) \quad (2.46)$$

Example

- (i) $f(z) = \frac{1}{z}$ is analytic at every point in the finite plane except $z = 0$.

- (ii) $f(z) = \bar{z}$ is nowhere analytic.
- (iii) Every polynomial is an entire function.

singular point is when it's analytic around z_0 but not at z_0

Definition Let $u(x, y)$ be a real-valued function of two variables x and y . If u is continuous on the entire domain of definition together with its first and second partial derivatives, and satisfies the **Laplace equation**

$$u_{xx}(x, y) + u_{yy}(x, y) = 0 \quad (2.60)$$

then u is called a **harmonic function**.

If a complex function $f = u + iv$ is analytic at a point, then the real part u and imaginary part v of f have continuous partial derivatives of all orders at that point (Theorem 4.4). Using this fact, we can prove Theorem 2.8.

Lemma Let $D \subset \mathbb{R}^2$ be an open connected set, $(a, b) \in D$, and $u : D \rightarrow \mathbb{R}$. If u is C^1 on D and one of the mixed second partial derivatives of u exists and is continuous at (a, b) , then the other mixed second partial derivative also exists at (a, b) and

$$\frac{\partial^2 u}{\partial y \partial x}(a, b) = \frac{\partial^2 u}{\partial x \partial y}(a, b).$$

Proof. Suppose u_{xx} exists on D and is continuous at (a, b) . Choose $r > 0$ such that $B_r(a, b) \subset D$, and for $|h|, |k| < \frac{r}{\sqrt{2}}$, let

$$\Delta u(h, k) = u(a + h, b + k) - u(a + h, b) - u(a, b + k) + u(a, b).$$

Applying the Mean Value Theorem twice, we can choose scalars $s, t \in (0, 1)$ such that

$$\Delta u(h, k) = k \frac{\partial u}{\partial y}(a + h, b + tk) - k \frac{\partial u}{\partial y}(a, b + tk) = hk \frac{\partial^2 u}{\partial x \partial y}(a + sh, b + tk).$$

Since the mixed partial $\frac{\partial^2 u}{\partial x \partial y}$ is continuous at (a, b) ,

$$\lim_{h \rightarrow 0} \lim_{k \rightarrow 0} \frac{\Delta u(h, k)}{\dots}$$

Complex Integral

from the reason that

$$(a + b)' = a' + b'$$

it happens to complex terms with same reason,

$$w'(t) = u'(t) + iv'(t)$$

with same happening to integral

$$\int [f(t) + g(t)] = \int f(t) + \int g(t)$$

and if it's intregable on $[a, b]$, it shows that

$$Re \int w(t)dt = \int Rew(t)dt.$$

and same for Im

Cauchy-Goursat

For f continuous on D with an antiderivative on D , for any closed contour C in D :

$$\int_C f(z) dz = 0.$$

Let $C : z = z(t), a \leq t \leq b$, be a positively oriented simple closed contour, f analytic on and inside C . :

$$\int_C f(z) dz = \int_a^b f[z(t)]z'(t) dt.$$

With $f(z) = u(x, y) + iv(x, y)$, $z(t) = x(t) + iy(t)$:

$$f[z(t)]z'(t) = u(x(t), y(t))x'(t) - v(x(t), y(t))y'(t) + i(u(x(t), y(t))y'(t) + v(x(t), y(t))x'(t)).$$

As a complex function of real t , by Definition

$$\int_C f(z) dz = \int_a^b (ux' - vy') dt + i \int_a^b (vx' + uy') dt.$$

In terms of real line integrals:

$$\int_C f(z) dz = \int_C u dx - v dy + i \int_C v dx + u dy.$$

follows formally by substituting $u+iv$ for $f(z)$ and $dx+idy$ for dz , expanding. Also holds for C any contour (not just simple closed) with $f[z(t)]$ piecewise continuous on C .

Let $R \subset R^2$ be a region in the plane whose boundary $C = \partial R$ is a piecewise smooth curve, positively (counterclockwise) oriented. If $P, Q : R \rightarrow R$ are continuously differentiable on R , then

$$\int_C P dx + Q dy = \iint_R (Q_x - P_y) dA.$$

Since f is analytic on R , f is continuous on R , so u, v are continuous on R . If f' is continuous on R , then u, v are continuously differentiable on R . By Green's theorem:

$$\int_C f(z) dz = \iint_R (-v_x - u_y) dA + i \iint_R (u_x - v_y) dA.$$

Since f is analytic, Cauchy–Riemann equations hold:

$$u_x = v_y, \quad u_y = -v_x.$$

So the integrands vanish identically on R , giving:

Let C be a simple closed contour and f analytic on the region R enclosed by C , with f' continuous on R . Then

$$\int_C f(z) dz = 0.$$

If the integral is 0, the orientation of C is immaterial. If C is clockwise,

$$\int_C f(z) dz = - \int_{-C} f(z) dz,$$

The hypothesis “ f' continuous” in Cauchy’s theorem was removed by Goursat.

Let C be a simple closed contour, f analytic on and inside C . Then

$$\int_C f(z) dz = 0.$$

Unless stated otherwise, simple closed contours C are positively (counterclockwise) oriented. Let R be C together with its interior. Partition R by lines parallel to the x - and y -axes, equally spaced, into *squares*; a square containing boundary points not in R has those points removed, the remainder called a *partial square*. R is covered by finitely many squares and partial squares.

Let f be analytic on the closed region R . For any $\varepsilon > 0$, R can be covered by finitely many squares and partial squares $\sigma_1, \dots, \sigma_n$, each containing a fixed point $z_j \in \sigma_j$ such that for all $z \in \sigma_j$,

$$\left| \frac{f(z) - f(z_j)}{z - z_j} - f'(z_j) \right| < \varepsilon \quad (z \neq z_j).$$

Step 1. Suppose, for contradiction, a square or partial square σ_j exists for which no point z_j satisfies for all other $z \in \sigma_j$.

If σ_j is a square, bisect each side to form 4 smaller squares. If σ_j is a partial square, subdivide similarly and discard pieces lying outside R . If among the resulting small squares one still has no z_j satisfying for all other points, subdivide further. Continuing this process for each subregion as needed, after finitely many repetitions R can be covered by finitely many squares/partial squares satisfying Lemma

Step 2. Suppose instead that after subdividing one of the original subregions indefinitely, no required points z_j ever exist — a contradiction is derived.

Denote by σ_0 the original subregion (square or, if partial, the entire square containing it). Repeated bisection gives an infinite nested sequence

$$\sigma_0, \sigma_1, \sigma_2, \dots, \sigma_{k-1}, \sigma_k, \dots \quad (4.52)$$

The diagonals shrink to 0, so there is a unique common point z_0 in all σ_k ; each contains points of R other than z_0 . Since the squares' diagonals shrink below any δ , every δ -neighborhood $|z - z_0| < \delta$ eventually contains some σ_k entirely, hence contains points of R other than z_0 . Thus z_0 is an accumulation point of R .

Since f is analytic on R , f is analytic at z_0 , so $f'(z_0)$ exists. By definition of the derivative, for each $\varepsilon > 0$ there is a δ -neighborhood $|z - z_0| < \delta$ such that for all $z \neq z_0$ in it,

$$\left| \frac{f(z) - f(z_0)}{z - z_0} - f'(z_0) \right| < \varepsilon.$$

Choosing K large enough that the diagonal of σ_K is less than δ gives a contradiction, completing the proof.

For C_j the boundary of σ_j (positively oriented), define on σ_j the error function

$$\delta_j(z) = \frac{f(z) - f(z_j)}{z - z_j} - f'(z_j) \quad (z \neq z_j), \quad \delta_j(z_j) = 0,$$

so $f(z) = f(z_j) + f'(z_j)(z - z_j) + (z - z_j)\delta_j(z)$, with $|\delta_j(z)| < \varepsilon$ on σ_j by (4.51). Integrating around C_j kills the linear part (it has an antiderivative), so

$$\int_{C_j} f(z) dz = \int_{C_j} (z - z_j)\delta_j(z) dz.$$

If σ_j is a square with side s_j , then $|z - z_j| < \sqrt{2}s_j$ for z on C_j , and the contour has length $4s_j$, giving

$$\left| \int_{C_j} (z - z_j)\delta_j(z) dz \right| < \sqrt{2}s_j \varepsilon \cdot 4s_j = 4\sqrt{2}A_j\varepsilon, \quad (4.59)$$

where $A_j = s_j^2$.

If σ_j is a partial square, with C_j consisting of part of C (length L_j) and part of the square boundary (length at most $4s_j$), then the bound becomes

$$\left| \int_{C_j} (z - z_j)\delta_j(z) dz \right| < \sqrt{2}s_j\varepsilon(4s_j + L_j) < 4\sqrt{2}A_j\varepsilon + \sqrt{2}SL_j\varepsilon,$$

where S bounds all the s_j .

Summing over all squares and partial squares: interior shared edges cancel (traversed oppositely), leaving only the original integral over C plus the sum over the partial squares' boundary segments along C , whose total length is C 's length L . The sum of all A_j does not exceed S^2 . C

$$\left| \int_C f(z) dz \right| < (4\sqrt{2}S^2 + \sqrt{2}SL)\varepsilon.$$

Since ε is arbitrary, the right side can be made as small as desired, so

$$\int_C f(z) dz = 0,$$

proving the Cauchy–Goursat theorem for a region enclosed by a simple closed contour. We now extend the theorem to a wider class of domains.

A domain D is *simply connected* if every simple closed contour in D can be continuously shrunk to a point within D . A connected domain that is not simply connected is called *multiply connected*.

The region enclosed by a simple closed contour is simply connected. An annulus bounded by two circles is not simply connected.

The Cauchy–Goursat theorem extends to simply connected domains:

If f is analytic throughout a simply connected domain D , then for every closed contour C lying in D ,

$$\int_C f(z) dz = 0.$$

If C is a simple closed contour, the result follows directly from the Cauchy–Goursat theorem. If C is a closed contour that intersects itself finitely many times, C can be decomposed into finitely many simple closed contours; applying the Cauchy–Goursat theorem to each piece gives the corresponding result for C . The case where the closed contour intersects itself infinitely many times requires a more detailed treatment, omitted here.

If f is analytic throughout a simply connected domain D , then f has an antiderivative in D . Follows from Theorems

Let D be a simply connected domain with $0 \notin D$. Then there exists an analytic function $F(z)$ satisfying $e^{F(z)} = z$. Here $F(z)$ is to be understood modulo $2\pi i$.

Cauchy Integral Formula and Applications

In this section we present another of Cauchy’s theorems.

Let f be analytic on and inside a positively oriented simple closed contour C . If z_0 is any point inside C , then

$$f(z_0) = \frac{1}{2\pi i} \int_C \frac{f(z)}{z - z_0} dz.$$

Since f is continuous at $z = z_0$, for given $\varepsilon > 0$ there exists $\delta > 0$ such that

$$|z - z_0| < \delta \implies |f(z) - f(z_0)| < \varepsilon.$$

Choose $\rho < \delta$ and let C_0 be the positively oriented circle $|z - z_0| = \rho$, chosen small enough to lie inside C . Then

$$|z - z_0| = \rho \implies |f(z) - f(z_0)| < \varepsilon.$$

The function $f(z)/(z - z_0)$ is analytic in the region between the contours C and C_0 , so by Corollary

$$\int_C \frac{f(z)}{z - z_0} dz = \int_{C_0} \frac{f(z)}{z - z_0} dz.$$

Subtracting $f(z_0) \int_{C_0} \frac{dz}{z - z_0}$ from both sides of this equation:

$$\int_C \frac{f(z)}{z - z_0} dz - f(z_0) \int_{C_0} \frac{dz}{z - z_0} = \int_{C_0} \frac{f(z) - f(z_0)}{z - z_0} dz.$$

Since

$$\begin{aligned} \int_{C_0} \frac{dz}{z - z_0} &= 2\pi i, \\ \int_C \frac{f(z)}{z - z_0} dz - 2\pi i f(z_0) &= \int_{C_0} \frac{f(z) - f(z_0)}{z - z_0} dz, \\ \left| \int_{C_0} \frac{f(z) - f(z_0)}{z - z_0} dz \right| &< \frac{\varepsilon}{\rho} \cdot 2\pi\rho = 2\pi\varepsilon. \\ \left| \int_C \frac{f(z)}{z - z_0} dz - 2\pi i f(z_0) \right| &< 2\pi\varepsilon. \end{aligned}$$

ε is an arbitrary positive number

Let C be the positively oriented circle $|z| = 2$. Evaluate

$$\int_C \frac{z}{(z^2 + 9)(z - i)} dz.$$

The integrand $z/[(z^2 + 9)(z - i)]$ is not analytic inside C , because of the point $z = i$. Hence the Cauchy–Goursat theorem cannot be applied directly. Among the zeros of the denominator, only $z_0 = i$ lies inside C . The function excluding this point,

$$f(z) = \frac{z}{z^2 + 9},$$

is analytic inside C , so the hypotheses of the Cauchy integral formula are satisfied. Rewriting the given integral as

$$\int_C \frac{f(z)}{z - i} dz,$$

the Cauchy integral formula (4.64) gives the value $2\pi i f(i) = -\frac{\pi}{4}$.

We now obtain an integral representation for $f'(z)$. For $0 < |\Delta z| < d$:

$$\frac{f(z + \Delta z) - f(z)}{\Delta z} = \frac{1}{2\pi i} \int_C \left(\frac{1}{s - z - \Delta z} - \frac{1}{s - z} \right) \frac{f(s)}{\Delta z} dz = \frac{1}{2\pi i} \int_C \frac{f(s)}{(s - z - \Delta z)(s - z)} ds,$$

where $d < \text{dist}(z, C) = \inf\{|z - s| : s \in C\}$. Using the continuity of f on C , we show that

$$\lim_{\Delta z \rightarrow 0} \frac{1}{2\pi i} \int_C \frac{f(s)}{(s - z - \Delta z)(s - z)} ds = \frac{1}{2\pi i} \int_C \frac{f(s)}{(s - z)^2} ds.$$

To this end we first compute

$$\int_C \left[\frac{1}{(s - z - \Delta z)(s - z)} - \frac{1}{(s - z)^2} \right] f(s) ds = \Delta z \int_C \frac{f(s)}{(s - z - \Delta z)(s - z)^2} ds.$$

Let $M = \max_{z \in C} |f(z)|$ and let L be the length of the contour C . Since $|s - z| > d$ and

$$|s - z - \Delta z| \geq ||s - z| - |\Delta z|| > d - |\Delta z|,$$

we have, as $|\Delta z| \rightarrow 0$,

$$\left| \Delta z \int_C \frac{f(s)}{(s - z - \Delta z)(s - z)^2} ds \right| \leq \frac{|\Delta z|ML}{(d - |\Delta z|)^2} \rightarrow 0.$$

Hence

$$\lim_{\Delta z \rightarrow 0} \frac{f(z + \Delta z) - f(z)}{\Delta z} = \frac{1}{2\pi i} \int_C \frac{f(s)}{(s - z)^2} ds,$$

and so finally

$$f'(z) = \frac{1}{2\pi i} \int_C \frac{f(s)}{(s - z)^2} ds.$$

Using the same method, we obtain

$$f''(z) = \frac{1}{\pi i} \int_C \frac{f(s)}{(s - z)^3} ds.$$

By induction, we obtain

$$f^{(n)}(z) = \frac{n!}{2\pi i} \int_C \frac{f(s)}{(s - z)^{n+1}} ds, \quad n = 0, 1, 2, \dots,$$

where $f^{(0)} = f$.

Equation asserts the existence of the second derivative of f at each point z inside C . In fact, if a function f is analytic at a point, then its derivative f' is also analytic at that point. This is because, if f is analytic at the point z , there exists a circle about z on and inside which f is analytic. $f''(z)$ exists at each point inside that circle, and hence f' is analytic at z . Applying the same argument to the analytic function f' shows that f'' is also analytic. Continuing in this way, we obtain the following theorem.

If a function f is analytic at a point z_0 , then all the derivatives $f^{(n)}$ of f are also analytic at z_0 .

However, Theorem does not hold for real functions. If

$$f(z) = u(x, y) + iv(x, y)$$

is analytic at the point $z = (x, y)$, then the analyticity of f' guarantees the continuity of f' . Since

$$f'(z) = u_x(x, y) + iv_x(x, y) = v_y(x, y) - iu_y(x, y),$$

the first-order partial derivatives of u, v are continuous at that point; that is, u, v are C^1 functions. Moreover, f'' is also analytic and continuous at z , with

$$f''(z) = u_{xx}(x, y) + iv_{xx}(x, y) = v_{yx}(x, y) - iu_{yx}(x, y),$$

and so on.

Let C be a positively oriented closed contour and z_0 an interior point of C . Then

$$\int_C \frac{1}{(z - z_0)^n} dz = \begin{cases} 2\pi i, & n = 1, \\ 0, & n \neq 1. \end{cases}$$

Setting $f(z) = 1$, follows immediately from the Cauchy integral formula

The proof of the next theorem relies on the fact that the derivative of an analytic function is itself analytic.

[Morera] If f is continuous throughout a domain D , and

$$\int_C f(z) dz = 0$$

Suppose the hypotheses of the theorem hold. Then f has an antiderivative F . Since $F' = f$, F is analytic. the derivative of an analytic function is itself analytic, so f is analytic.

If D is a simply connected domain, Morera's theorem, applied to the set of continuous functions on D , states the converse of the Cauchy-Goursat theorem.

[Weierstrass] Suppose the sequence of functions $f_n(z)$ is analytic on a domain Ω_n , and the sequence $\{f_n(z)\}$ converges to a limit function $f(z)$ on Ω , uniformly on every compact subset of Ω . Then $f(z)$ is analytic on Ω . Moreover, $f'_n(z)$ converges uniformly to $f'(z)$ on every compact subset of Ω .

1. Let $|z - a| \leq r$ be a closed disk contained in Ω . By hypothesis, this disk is contained in Ω_n for all n greater than some n_0 . If γ is any closed curve contained in $|z - a| < r$, then by Cauchy's theorem, for $n > n_0$,

$$\int_{\gamma} f_n(z) dz = 0.$$

Using uniform convergence on γ ,

$$\int_{\gamma} f(z) dz = \lim_{n \rightarrow \infty} \int_{\gamma} f_n(z) dz = 0,$$

and by Morera's theorem, $f(z)$ is analytic on $|z - a| < r$. Hence $f(z)$ is analytic throughout Ω .

2. Since f_n is analytic, by the Cauchy integral formula,

$$f_n(z) = \frac{1}{2\pi i} \int_C \frac{f_n(\zeta)}{\zeta - z} d\zeta,$$

where C is the circle $|\zeta - a| = r$ and $|z - a| < r$. As $n \rightarrow \infty$, by uniform convergence,

$$f(z) = \frac{1}{2\pi i} \int_C \frac{f(\zeta)}{\zeta - z} d\zeta,$$

and this formula shows that $f(z)$ is analytic on the disk. From the differentiation formula

$$f'_n(z) = \frac{1}{2\pi i} \int_C \frac{f_n(\zeta)}{(\zeta - z)^2} d\zeta,$$

the same method gives

$$\lim_{n \rightarrow \infty} f'_n(z) = \frac{1}{2\pi i} \int_C \frac{f(\zeta)}{(\zeta - z)^2} d\zeta = f'(z),$$

and it is easy to show that this convergence is uniform on $|z - a| \leq \rho < r$. Any compact subset of Ω can be covered by finitely many such closed disks, so the convergence is uniform on every compact subset.

Continuing the proof process above, for each positive integer k , $f_n^{(k)}(z)$ converges uniformly to $f^{(k)}(z)$ on every compact subset of Ω .

[Laurent] Let f be analytic in the annular domain $R_1 < |z - z_0| < R_2$. Let C be any simple closed contour about z_0 , oriented positively, lying in this domain. Then at each point z in the domain,

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n + \sum_{n=1}^{\infty} \frac{b_n}{(z - z_0)^n} \quad (R_1 < |z - z_0| < R_2)$$

where

$$a_n = \frac{1}{2\pi i} \int_C \frac{f(z)}{(z - z_0)^{n+1}} dz \quad (n = 0, 1, 2, \dots)$$

and

$$b_n = \frac{1}{2\pi i} \int_C \frac{f(z)}{(z - z_0)^{-n+1}} dz \quad (n = 1, 2, \dots)$$

Equation can be written compactly as

$$f(z) = \sum_{n=-\infty}^{\infty} c_n (z - z_0)^n \quad (R_1 < |z - z_0| < R_2)$$

where

$$c_n = \frac{1}{2\pi i} \int_C \frac{f(z)}{(z - z_0)^{n+1}} dz \quad (n = 0, \pm 1, \pm 2, \dots)$$

The series expressions are called the Laurent Series
[Uniqueness of Laurent series] If a series

$$\sum_{n=-\infty}^{\infty} c_n(z-z_0)^n = \sum_{n=0}^{\infty} a_n(z-z_0)^n + \sum_{n=1}^{\infty} \frac{b_n}{(z-z_0)^n}$$

converges to $f(z)$ at every point of some annular domain about z_0 , then this is the Laurent series for f at $z = z_0$.

Applying Theorem to the doubly infinite series

$$f(z) = \sum_{m=-\infty}^{\infty} c_m(z-z_0)^m$$

gives

$$\int_C g(z)f(z) dz = \sum_{m=-\infty}^{\infty} c_m \int_C g(z)(z-z_0)^m dz$$

where C is a circle in the annular domain. With

$$g(z) = \frac{1}{2\pi} \cdot \frac{1}{(z-z_0)^{n+1}} \quad (n = 0, \pm 1, \pm 2, \dots)$$

we obtain

$$\int_C \frac{f(z)}{(z-z_0)^{n+1}} dz = \sum_{m=-\infty}^{\infty} c_m \int_C (z-z_0)^{n+m+1} dz$$

and, for all integers m, n ,

$$\frac{1}{2\pi i} \int_C \frac{dz}{(z-z_0)^{n-m+1}} = \begin{cases} 0, & m \neq n \\ 1, & m = n \end{cases}$$

From which

$$\sum_{m=0}^{\infty} a_m \int_C g(z)(z-z_0)^m dz = c_n.$$

therefore the equation is the Laurent series for f at the point z_0 .

The following property is one possessed only by complex functions.

[Residue Theorem] Let C be a positively oriented simple closed contour. If f is analytic on and inside C except at finitely many singular points $z_1, \dots, z_n$ inside C , then

$$\int_C f(z) dz = 2\pi i \sum_{k=1}^n \text{Res}(f; z = z_k).$$

Let z_k ($k = 1, \dots, n$) be centers of small positively oriented circles C_k , contained in the interior of C , chosen small enough to be pairwise disjoint. Then f

is analytic on the region bounded by $C, C_1, \dots, C_n$, and its interior is multiply connected. Applying the Cauchy–Goursat theorem,

$$\int_C f(z) dz - \sum_{k=1}^n \int_{C_k} f(z) dz = 0.$$

Since

$$\int_{C_k} f(z) dz = 2\pi i \sum_{k=1}^n \text{Res}(f; z = z_k),$$

[Argument Principle] Let $f(z)$ be meromorphic in the interior of a positively oriented simple closed contour C , analytic and nonzero on C . If f has Z zeros and P poles inside C (counted with multiplicity/order), then

$$\frac{1}{2\pi} \Delta_C \arg f(z) = Z - P.$$

Let $C : z = z(t)$, $a \leq t \leq b$, be a parametrization. We compute $\int_C \frac{f'(z)}{f(z)} dz$ two ways.

Method 1.

$$\int_C \frac{f'(z)}{f(z)} dz = \int_a^b \frac{f'(z(t))z'(t)}{f(z(t))} dt.$$

Since $\Gamma = f(C)$ does not pass through the origin, write

$$\Gamma : f(z(t)) = \rho(t)e^{i\phi(t)}, \quad a \leq t \leq b.$$

Then

$$f'(z(t))z'(t) = \frac{d}{dt}f(z(t)) = \frac{d}{dt}\rho(t)e^{i\phi(t)} = \rho'(t)e^{i\phi(t)} + i\rho(t)e^{i\phi(t)}\phi'(t).$$

Hence

$$\int_C \frac{f'(z)}{f(z)} dz = \int_a^b \frac{\rho'(t)}{\rho(t)} dt + i \int_a^b \phi'(t) dt = [\ln \rho(t)]_a^b + i[\phi(t)]_a^b.$$

Since $\rho(a) = \rho(b)$ and $\phi(b) - \phi(a) = \Delta_C \arg f(z)$,

$$\int_C \frac{f'(z)}{f(z)} dz = i \Delta_C \arg f(z).$$

Method 2. The integrand $f'(z)/f(z)$ is analytic inside C except at the zeros and poles of f .

If z_0 is a zero of f of order m_0 , then $f(z) = (z - z_0)^{m_0}g(z)$ with g analytic and $g(z_0) \neq 0$. Then

$$f'(z) = m_0(z - z_0)^{m_0-1}g(z) + (z - z_0)^{m_0}g'(z),$$

so

$$\frac{f'(z)}{f(z)} = \frac{m_0}{z - z_0} + \frac{g'(z)}{g(z)}.$$

Since g'/g is analytic at z_0 , it has a Taylor series there, type reasoning z_0 is a simple pole of f'/f with

$$\operatorname{Res}\left(\frac{f'(z)}{f(z)}; z = z_0\right) = m_0.$$

If z_1 is a pole of f of order m_p , then

$$f(z) = (z - z_1)^{-m_p} \phi(z),$$

with ϕ analytic and $\phi(z_1) \neq 0$. with m_0 replaced by $-m_p$, so z_1 is a simple pole of f'/f with

$$\operatorname{Res}\left(\frac{f'(z)}{f(z)}; z = z_1\right) = -m_p.$$

By the Residue Theorem,

$$\int_C \frac{f'(z)}{f(z)} dz = 2\pi i(Z - P).$$

therefore

$$i \Delta_C \arg f(z) = 2\pi i(Z - P) \implies \frac{1}{2\pi} \Delta_C \arg f(z) = Z - P.$$

6.3 Methods for Finding Residues

I. Finding Residues at Poles

Suppose f has a pole of order m at z_0 . Then in a punctured neighborhood $0 < |z - z_0| < R_2$, f has Laurent expansion

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n + \frac{b_1}{z - z_0} + \frac{b_2}{(z - z_0)^2} + \cdots + \frac{b_m}{(z - z_0)^m}, \quad b_m \neq 0.$$

Define

$$\phi(z) = \begin{cases} (z - z_0)^m f(z), & z \neq z_0 \\ b_m, & z = z_0 \end{cases}$$

Then on $|z - z_0| < R_2$,

$$\phi(z) = b_m + b_{m-1}(z - z_0) + \cdots + b_2(z - z_0)^{m-2} + b_1(z - z_0)^{m-1} + \sum_{n=0}^{\infty} a_n (z - z_0)^{n+m}.$$

So ϕ is analytic on $|z - z_0| < R_2$, in particular at $z = z_0$, with $\phi(z_0) = b_m \neq 0$. Hence

$$f(z) = \frac{\phi(z)}{(z - z_0)^m},$$

with ϕ analytic at z_0 and $\phi(z_0) \neq 0$.

Conversely, suppose $f(z)$ can be written in the form with ϕ analytic on $|z - z_0| < \varepsilon$. Then ϕ has a Taylor expansion

$$\phi(z) = \phi(z_0) + \phi'(z_0)(z - z_0) + \frac{\phi''(z_0)}{2!}(z - z_0)^2 + \cdots + \frac{\phi^{(m-1)}(z_0)}{(m-1)!}(z - z_0)^{m-1} + \sum_{n=m}^{\infty} \frac{\phi^{(n)}(z_0)}{n!}(z - z_0)^n.$$

for $0 < |z - z_0| < \varepsilon$,

$$f(z) = \frac{\phi(z_0)}{(z - z_0)^m} + \frac{\phi'(z_0)}{(z - z_0)^{m-1}} + \frac{\phi''(z_0)/2!}{(z - z_0)^{m-2}} + \cdots + \frac{\phi^{(m-1)}(z_0)/(m-1)!}{z - z_0} + \sum_{n=m}^{\infty} \frac{\phi^{(n)}(z_0)}{n!}(z - z_0)^{n-m}.$$

Since $\phi(z_0) \neq 0$, f has a pole of order m at z_0 . The residue of f at z_0 is

$$\text{Res}(f; z = z_0) = \begin{cases} \phi(z_0), & m = 1 \\ \frac{\phi^{(m-1)}(z_0)}{(m-1)!}, & m \geq 2 \end{cases}$$